Johannes Solibieda 261295737

Fixed Income Assignment Part 1 Report

The goal of this project was to replicate the three-step BDT short-rate tree provided in the Excel example and to estimate smooth yield and volatility term structures using the Nelson–Siegel model. Both tasks involve nonlinear calibration using fminsearch() and observed market inputs. The general approach is to construct model-implied bond prices and fitted yield/volatilities to adjust model parameters until the model matches the data as closely as possible.

1. Replicating the BDT excel example in Matlab

The idea of this subtask was to construct a short rate tree from observed bond prices and volatilities on the market using a binominal bond pricing model based on yields and volatility of rates. The model price and volatility need to match the observed data, after which we can extract the yield and volatility of the bond tree and build the short rate tree.
In a first step, all parameters were set, such as timesteps and observed market data. In a second step, the short rate for the shortest timespan was calculated using a simple fminsearch for the price from which the yield, respectively the rate can be inferred.
For further timespans, volatility and base rate (lowest short rate) needed to be calculated using a loop function and calibration of both parameters at the same time. This was done by constructing a placeholder bond tree with corresponding size which was induced/calculated by discounting backwards using placeholder rates for the calibration year and already set rates for the prior years. The placeholder rates were then chosen by fminsearch such that the difference to the observed market data is minimized (they should be equal). From that, the short base rate and volatility can be calculated, and the short rate tree can be completed (see visual 1 below).

```
The short tree is:
    0.0953       NaN       NaN
    0.0923    0.1350       NaN
    0.0910    0.1284    0.1810
```

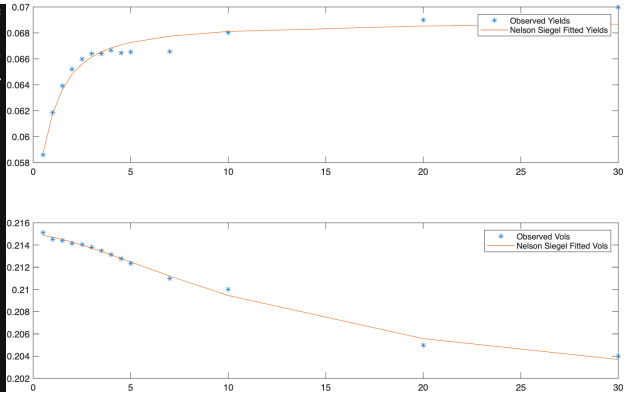
Visual 1: resulting short rate tree

2. Estimating Yield and Volatility Term Structures

The idea of this subtask was to apply the Nelson Siegel YIELD function to an incomplete (in terms of maturities) set of market yields and volatilities as a preparation to create a short rate tree from the data to value complex securities such as mortgage backed security, where changes in the short rate have a big influence on the value of the security (through prepayment option and discounting). In the first step, a squared error function was set up calculating the residual of true and fitted values of the Nelson Siegel curve using placeholder parameters. In the second step, the placeholder parameters were optimized using the fminsearch function to minimize the squared error. This was done twice, once for the yield curve and once for the volatility curve. Afterwards, the data is plotted to visualize the curves and fitted values (see visual 2 below).
We see that the volatility curve fits very well while the yield curve does not seem to visually fit as well as the volatility curve. This could be overcome by using more complicated smoother functions but might also introduce overfitting.

```
The fitted data table is:
    m        y         vol      nelsonSiegelYield    nelsonSiegelVol
    ___    _______    _______    _________________    _______________

    0.5      0.0586     0.2151        0.058699            0.21487
      1    0.061838    0.21452        0.061794             0.2147
    1.5    0.063908    0.21438        0.063649             0.2145
      2    0.065198    0.21416         0.06482            0.21426
    2.5     0.06597    0.21403          0.0656              0.214
      3      0.0664    0.21379        0.066144            0.21372
    3.5    0.066386    0.21349        0.066541            0.21342
      4    0.066661    0.21315        0.066841            0.21311
    4.5    0.066472     0.2128        0.067076             0.2128
      5     0.06654    0.21236        0.067264            0.21248
      7     0.06655      0.211        0.067747            0.21121
     10       0.068       0.21        0.068109            0.20946
     20       0.069      0.205        0.068532            0.20558
     30        0.07      0.204        0.068673             0.2037
```



Visual 2: fitted data table and yield/volatility curve plots